

Capstone Proposal

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News Summariser

Overview :

A serverless news analysis platform that fetches daily tech articles from NewsAPI, enriches them with Python-based NLP analysis (sentiment, entities, key phrases), and stores results in DynamoDB for historical tracking. Fully serverless using AWS Lambda with automated 24-hour scheduling via EventBridge.

The News Analyzer is a fully serverless cloud application that automates the collection, enrichment, and analysis of technology news articles. By leveraging AWS Lambda and managed services, the platform provides a cost-effective, scalable infrastructure for processing real-world data from the NewsAPI external service. The whole news summariser will be shown in a form of web app.

Services:

Service	Role	What It Does
ECS Fargate	Compute	Runs your containerized Python app that fetches & summarizes news
ECR	Container Registry	Stores your Docker image
EventBridge	Trigger/Scheduler	Runs the ECS task every 24 hours (web app)
DynamoDB	Persistent State	Stores summarized articles with dates
Secrets Manager	Secret Management	Safely stores NewsAPI key
CloudWatch	Observability	Logs, metrics, monitoring

Architecture Diagram :

